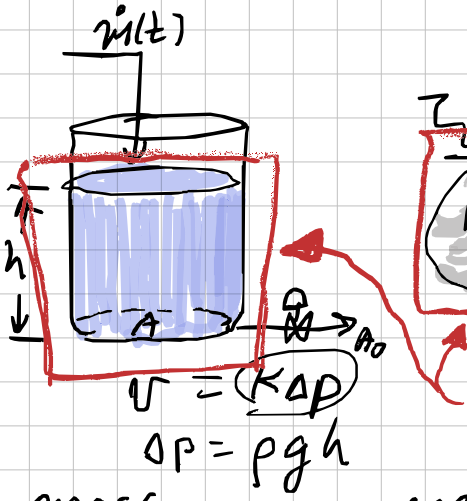
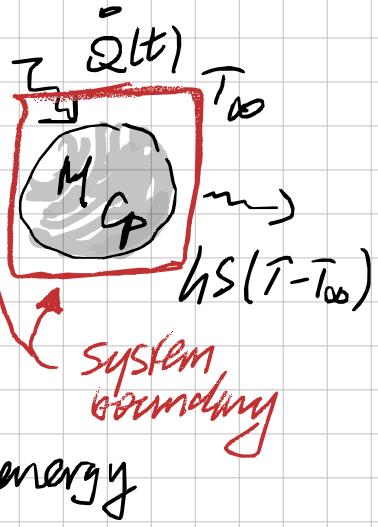
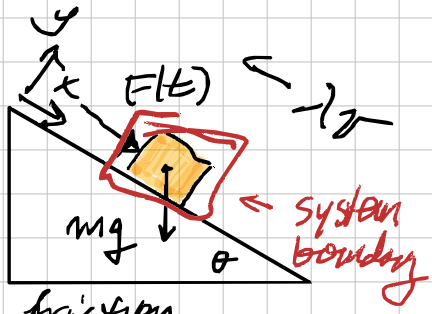
Differential balances and differential equations



mass



energy



friction

$f = \mu N$

momentum / force

balance equation:

accumulation = in − out + generation

$t \rightarrow t + \Delta t$

mass

$$m(t+\Delta t) - m(t) = \dot{m}(t)\Delta t - \rho v A_0 \Delta t$$

$$\rho A h(t+\Delta t) - \rho A h(t) = \dot{m}(t)\Delta t - \rho A_0 K \rho h \Delta t$$

$$\rho A \frac{dh}{dt} = \dot{m}(t) - \rho A_0 K \rho h(t)$$

energy

$$U(t+\Delta t) - U(t) =$$

momentum

$$P_x(t+\Delta t) - P_x(t) =$$

$$\rho A \frac{dh}{dt} = \dot{m}_i(t) - \rho A_o K_p h(t)$$

$$\frac{dh}{dt} + \frac{A_o}{A} K_p h(t) = \dot{m}_i(t)$$

$$\frac{dy(t)}{dt} + a(t)\, y(t) = f(t)$$

$$\Delta U = m C_p \Delta T$$

energy

$$U(t+\Delta t) - U(t) = \dot{Q}(t)\,\Delta t - hS(T - T_\infty)\,\Delta t$$

$$mC_p\left[T(t+\Delta t) - T(t)\right] = \dot{Q}(t)\Delta t - hS(T - T_\infty)\Delta t$$

$$mC_p \frac{dT}{dt} = \dot{Q}(t) - hS(T - T_\infty)$$

$$\frac{dT}{dt} = \frac{\dot{Q}(t)}{mC_p} - \frac{hS}{mC_p}(T - T_\infty)$$

$$\frac{dT}{dt} + \frac{hS}{mC_p} T = \frac{\dot{Q}(t)}{mC_p} + \frac{hS}{mC_p} T_\infty$$

momentum

$$P_x(t+\Delta t) - P_x(t)$$
$$= F(t)\,\Delta t - F_{drag}\,\Delta t$$
$$= F(t)\,\Delta t - \lambda v\,\Delta t$$

$$\frac{dP}{dt} = F(t) - \lambda v$$

$$F = ma$$
$$= m\frac{dv}{dt}$$
$$F = \frac{d(mv)}{dt}$$
$$F = \frac{dP}{dt}$$
$$\Delta P = F\,\Delta t$$
$$P(t+\Delta t) - P(t) = F\,\Delta t$$

momentum

$$p_x(t+\Delta t) - p_x(t)$$
$$= F(t)\Delta t - F_{drag}\Delta t$$
$$= F(t)\Delta t - \lambda v \Delta t$$

$$\frac{dp_x}{dt} = F(t) - \lambda v_x \qquad p_x = m v_x$$

$$m\frac{dv_x}{dt} = F(t) - \lambda v_x$$

$$\frac{dv_x}{dt} + \frac{\lambda}{m} v_x = \frac{F(t)}{m}$$

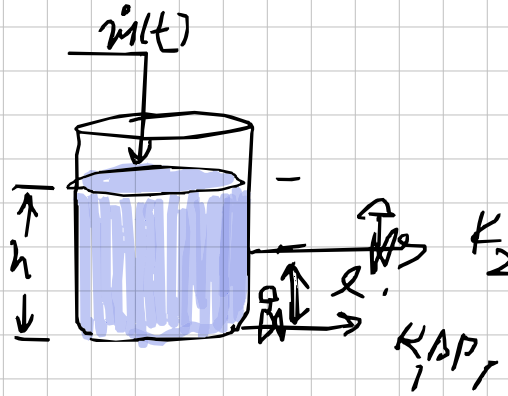
$$\int dx\, x^n = ? \qquad \int dt\, t^{1/\gamma} = ?$$

x^n $\qquad$ $t^{1/\gamma}$

$x \leftrightarrow t$

$n \longleftrightarrow 1/\gamma$

$$\frac{x^{n+1}}{n+1} \qquad \frac{t^{1/\gamma+1}}{1/\gamma+1}$$

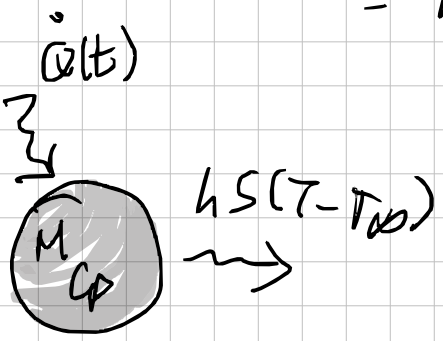


Tank with varying cross-sectional area



$$m(t+\Delta t) - m(t) = \dot{m}(t)\Delta t - \rho V_1 A_1 \Delta t - \rho V_2 A_2 \Delta t$$

$$= \dot{m}(t)\Delta t - \rho K_1 \rho g h A_1 \Delta t - \rho K_2 \rho g (h-l) A_2 \Delta t$$



$$\Delta m = \rho A \Delta h \quad \longleftrightarrow \quad \boxed{\Delta m = \rho A(h)\Delta h}$$

$$\Delta U = m C_p \Delta T \quad \longleftrightarrow \quad \Delta U = m C_p(T)\Delta T$$

$$m(t+\Delta t) - m(t) = -\rho K \rho g h \Delta t$$

$$\frac{dm}{dt} = -\rho K \rho g h$$

$$A(h) = \beta h$$

$$\rho A(h) \frac{dh}{dt} = -\rho K \rho g h$$

$$\frac{dh}{dt} = -\frac{K \rho g h}{A(h)}$$